

Linearized fluxes inside shell sector for 1D, 2D and 3D droplets. We solve the Reaction-Diffusion equation in the shell sector

```
In[ ]:= $Assumptions = B > 0 && Diff > 0 && cEqOut > 0 && R > 0 && L > 0 && l > 0
Out[ ]:= B > 0 && Diff > 0 && cEqOut > 0 && R > 0 && L > 0 && l > 0
```

Solve the stationary state equation for 1D droplets

```
In[ ]:= sol1D[x_] = FullSimplify[
  c[x] /. First@DSolve[{0 == Diff * Laplacian[c[x], {x}, "Cartesian"] + A - B * c[x],
    c[R] == cEqOut, c[R + L] == cfar}, c, x] /. Diff -> B * l^2]
Out[ ]:= 
$$\frac{e^{\frac{x}{l}} \left( -1 + \coth\left[\frac{l}{l}\right] \right) \left( A \sinh\left[\frac{l}{l}\right] + (A - B \text{cfar}) \sinh\left[\frac{R-x}{l}\right] + (-A + B \text{cEqOut}) \sinh\left[\frac{L+R-x}{l}\right] \right)}{B}$$

```

Check the boundary conditions

```
In[ ]:= FullSimplify[sol1D[R]]
Out[ ]:= cEqOut

In[ ]:= FullSimplify[sol1D[R + L]]
Out[ ]:= cfar
```

Check the solution by plugging it in the Helmholtz equation

```
In[ ]:= FullSimplify[Diff * Laplacian[sol1D[x], {x}, "Cartesian"] + A - B * sol1D[x]] /.
  Diff -> B * l^2
Out[ ]:= 0
```

Calculate line integrated fluxes at the droplet surface

```
In[ ]:= FluxInBound1D = FullSimplify[FullSimplify[-2 * Diff * D[sol1D[x], x] /. x -> R]]
Out[ ]:= 
$$- \frac{2 \text{Diff} \left( -A + B \text{cfar} + (A - B \text{cEqOut}) \cosh\left[\frac{l}{l}\right] \right) \text{Csch}\left[\frac{l}{l}\right]}{B l}$$

```

```
In[ ]:= FortranForm[FluxInBound1D]
```

```
Out[ ]:=FortranForm=
```

```
(-2*Diff*(-A + B*cFar + (A - B*cEqOut)*Cosh(L/l))*Csch(L/l))/(B*l)
```

Solve the stationary state equation for 2D droplets

```
In[ ]:= solI2D[r_] = FullSimplify[
```

```
c[r] /. First@DSolve[{0 == Diff*Laplacian[c[r], {r, ϕ}, "Polar"] + A - B*c[r],  
c[R] == cEqOut, c[R + L] == cFar}, c, r] /. Diff -> B*l^2]
```

```
Out[ ]:= 
$$\left( \pi \left( \text{BesselI}\left[0, \frac{L+R}{l}\right] \left( (-A + B \text{cEqOut}) \text{BesselY}\left[0, -\frac{i r}{l}\right] + A \text{BesselY}\left[0, -\frac{i R}{l}\right] \right) + \right. \right. \\ \left. \text{BesselI}\left[0, \frac{R}{l}\right] \left( (A - B \text{cFar}) \text{BesselY}\left[0, -\frac{i r}{l}\right] - A \text{BesselY}\left[0, -\frac{i (L+R)}{l}\right] \right) + \right. \\ \left. \text{BesselI}\left[0, \frac{r}{l}\right] \left( (-A + B \text{cFar}) \text{BesselY}\left[0, -\frac{i R}{l}\right] + (A - B \text{cEqOut}) \text{BesselY}\left[0, -\frac{i (L+R)}{l}\right] \right) \right) / \\ \left( 2 \left( -B \text{BesselI}\left[0, \frac{L+R}{l}\right] \text{BesselK}\left[0, \frac{R}{l}\right] + B \text{BesselI}\left[0, \frac{R}{l}\right] \text{BesselK}\left[0, \frac{L+R}{l}\right] \right) \right)$$

```

```
In[ ]:= sol2D[r] = FullSimplify[solI2D[r] /. BesselY[0, -I * x_] -> -2 / π * BesselK[0, x]]
```

```
Out[ ]:= 
$$\left( \text{BesselI}\left[0, \frac{L+R}{l}\right] \left( (-A + B \text{cEqOut}) \text{BesselK}\left[0, \frac{r}{l}\right] + A \text{BesselK}\left[0, \frac{R}{l}\right] \right) + \right. \\ \left. \text{BesselI}\left[0, \frac{R}{l}\right] \left( (A - B \text{cFar}) \text{BesselK}\left[0, \frac{r}{l}\right] - A \text{BesselK}\left[0, \frac{L+R}{l}\right] \right) + \right. \\ \left. \text{BesselI}\left[0, \frac{r}{l}\right] \left( (-A + B \text{cFar}) \text{BesselK}\left[0, \frac{R}{l}\right] + (A - B \text{cEqOut}) \text{BesselK}\left[0, \frac{L+R}{l}\right] \right) \right) / \\ \left( B \text{BesselI}\left[0, \frac{L+R}{l}\right] \text{BesselK}\left[0, \frac{R}{l}\right] - B \text{BesselI}\left[0, \frac{R}{l}\right] \text{BesselK}\left[0, \frac{L+R}{l}\right] \right)$$

```

Check the boundary conditions

```
In[ ]:= FullSimplify@sol2D[R]
```

```
Out[ ]:= sol2D[R]
```

```
In[ ]:= FullSimplify@sol2D[R + L]
```

```
Out[ ]:= sol2D[L + R]
```

Check the solution by plugging it in the Helmholtz equation

```
In[ ]:= FullSimplify[Diff*Laplacian[sol2D[r], {r, ϕ}, "Polar"] + A - B*sol2D[r]] /.  
Diff -> B*l^2
```

```
Out[ ]:= 0
```

Calculate perimeter integrated fluxes at the droplet surface

In[]:= **FluxInBound2D** = FullSimplify[FullSimplify[-2 π * R * Diff * D[sol2D[r], r] /. r $\rightarrow$ R]]

Out[]:=
$$\left(2 \text{Diff} \pi \left((A - B \text{cFar}) l + (-A + B \text{cEqOut}) R \right. \right. \\ \left. \left. \left(\text{BesselI}\left[1, \frac{R}{l}\right] \text{BesselK}\left[0, \frac{L+R}{l}\right] + \text{BesselI}\left[0, \frac{L+R}{l}\right] \text{BesselK}\left[1, \frac{R}{l}\right] \right) \right) \right) / \\ \left(B l \left(\text{BesselI}\left[0, \frac{L+R}{l}\right] \text{BesselK}\left[0, \frac{R}{l}\right] - \text{BesselI}\left[0, \frac{R}{l}\right] \text{BesselK}\left[0, \frac{L+R}{l}\right] \right) \right)$$

In[]:= **FortranForm[FluxInBound2D]**

Out[]//FortranForm=

$$(2 * \text{Diff} * \pi * ((A - B * \text{cFar}) * l + (-A + B * \text{cEqOut}) * R * \\ - (\text{BesselI}(1, R/l) * \text{BesselK}(0, (L + R)/l) + \text{BesselI}(0, (L + R)/l) * \text{BesselK}(1, R/l)) \\ - (B * l * (\text{BesselI}(0, (L + R)/l) * \text{BesselK}(0, R/l) - \text{BesselI}(0, R/l) * \text{BesselK}(0, (L + R)/l)))) /$$

Solve the stationary state equation for 3D droplets

In[]:= **sol3D[r_] = FullSimplify[c[r] /. First@DSolve[{0 == Diff * Laplacian[c[r], {r, ϕ , θ }, "Spherical"] + A - B * c[r],**

c[R] == cEqOut, c[R + L] == cFar}, c, r] /. Diff $\rightarrow$ B * l^2]

Out[]:=
$$\frac{1}{B r} e^{\frac{L}{l}} \left(-1 + \text{Coth}\left[\frac{L}{l}\right] \right) \\ \left(A r \text{Sinh}\left[\frac{L}{l}\right] - (A - B \text{cFar}) (L + R) \text{Sinh}\left[\frac{r - R}{l}\right] + (-A + B \text{cEqOut}) R \text{Sinh}\left[\frac{L - r + R}{l}\right] \right)$$

Check the boundary conditions

In[]:= **FullSimplify@sol3D[R]**

Out[]:= **cEqOut**

In[]:= **FullSimplify@sol3D[R + L]**

Out[]:= **cFar**

Check the solution by plugging it in the Helmholtz equation

In[]:= **FullSimplify[Diff * Laplacian[sol3D[r], {r, ϕ , θ }, "Spherical"] + A - B * sol3D[r] /. Diff $\rightarrow$ B * l^2]**

Out[]:= **0**

Calculate surface area integrated fluxes at the droplet surface

In[*]:= **FluxInBound3D =**

FullSimplify[FullSimplify[-4 π * R² * Diff * D[sol3D[r], r] /. r $\rightarrow$ R]]

Out[*]=
$$\frac{4 \text{Diff} \pi R \left(- \left((A - B \text{cEqOut}) \left(l + R \text{Coth} \left[\frac{L}{l} \right] \right) \right) + (A - B \text{cfar}) (L + R) \text{Csch} \left[\frac{L}{l} \right] \right)}{B l}$$

In[*]:= **FortranForm[FluxInBound3D]**

Out[*]//FortranForm=

$$(4 * \text{Diff} * \pi * R * (- ((A - B * \text{cEqOut}) * (l + R * \text{Coth}(L/l))) + (A - B * \text{cfar}) * (L + R) * \text{Csch}(L/l))$$
